

Exam Date & Time: 04-Jan-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.Tech. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS -
DECEMBER 2024 / JANUARY 2025

SUBJECT: ECE 2121/ECE_2121- ANALOG ELECTRONIC CIRCUITS
(MAKEUP)

ANALOG ELECTRONIC CIRCUITS [ECE 2121]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

Instructions to candidates:

Missing data may be suitably assumed.

- 1A) Design the common-gate stage for the circuit shown in Fig. Q1A for $V_{out}/V_{in} = 5$, $R_S = 0$, $R_3 = 500\Omega$, $R_{IN} = 50\Omega$, power budget = 2 mW, $V_{DD} = 1.8$ V. Assume $\mu_n C_{ox} = 100 \mu A/V^2$, $V_{TH} = 0.5$ V, and $\lambda = 0$. Assume the drain current to be 99.5% of the total current drawn by the circuit.

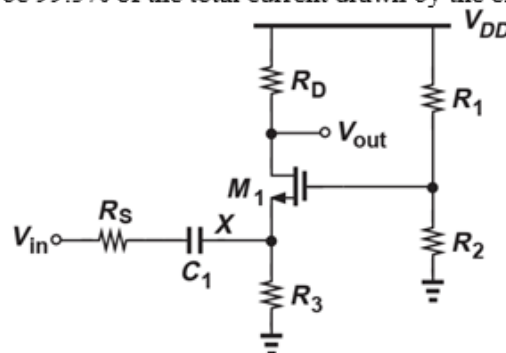


Fig. Q1A

- 1B) An NMOS device carries 1 mA with $V_{GS} - V_{TH} = 0.6$ V and 1.6 mA with $V_{GS} - V_{TH} = 0.8$ V. If the device operates in the triode region, calculate V_{DS} and W/L . Assume $\mu_n C_{ox} = 100 \mu A/V^2$, $\lambda = 0$. (3)
- 1C) Determine the voltage gain for the circuit shown in Fig. Q1C. Consider $\lambda = 0$.

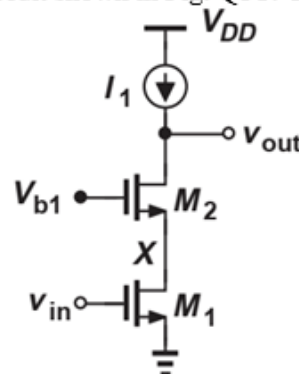
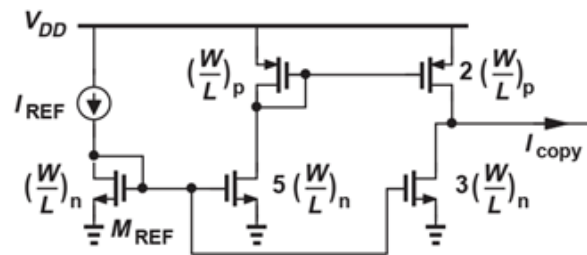


Fig. Q1C

- 2A) (4)

The diagram shows a differential amplifier circuit. It consists of two NMOS transistors, M_1 and M_2 , whose sources are connected to a common source node. This node is connected to ground through a current source I_{SS} . The gates of both transistors are connected to a common-mode input signal v_{in2} . The gates are also connected to a feedback network consisting of two resistors R_D in series with a voltage divider. The output voltage V_{out} is taken from the drain of M_1 . The circuit is powered by a supply voltage V_{DD} .

2B) Calculate I_{copy} in the circuit shown in Fig. Q2B, if the $I_{REF} = 1\text{mA}$. Assume all of the transistors operate in saturation.



(3)

(3)

(4)

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For the circuit shown in Fig. Q3B, find the pole frequencies and sketch the frequency response. Assume $\lambda = 0$, $g_m = 0.002\Omega^{-1}$, $C_{GD} = 90\text{fF}$, $C_{GS} = 200\text{fF}$, $C_{SB} = 120\text{fF}$, $V_{DD} = 1.8\text{V}$, $R_S = 200\Omega$, $R_L = 1\text{K}\Omega$, $C_L = 100\text{fF}$, $I = 1\text{mA}$.

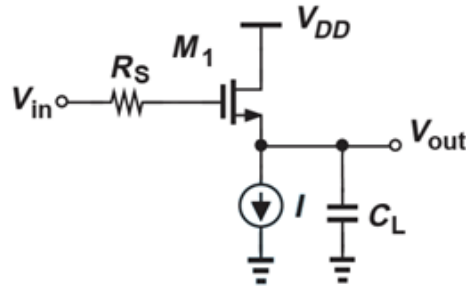


Fig. Q3B

- 3C) For the amplifier shown in Fig. Q3C, $g_m = (100\Omega)^{-1}$, $\lambda = 0$, $R_D = 2\text{K}\Omega$, $R_S = 200\Omega$, $C_{in} = 1\mu\text{F}$, and $C_L = 50\text{nF}$. Neglecting the MOSFET capacitance, sketch the frequency response of the amplifier indicating the pole frequencies.

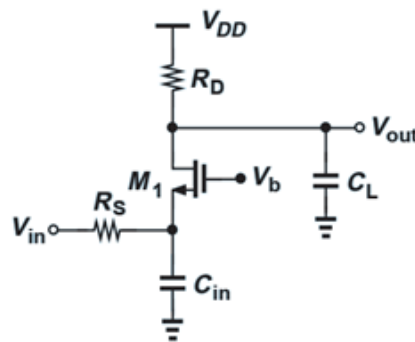


Fig. Q3C

- 4A) For the circuit shown in Fig. 4A, find the input and output poles at high frequencies for $\lambda = 0$. Sketch the frequency response.

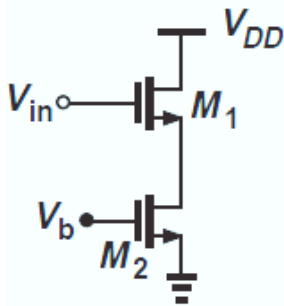


Fig. 4A

- 4B) Determine the polarity of feedback for the circuits shown in Fig. 4B. And justify the same.

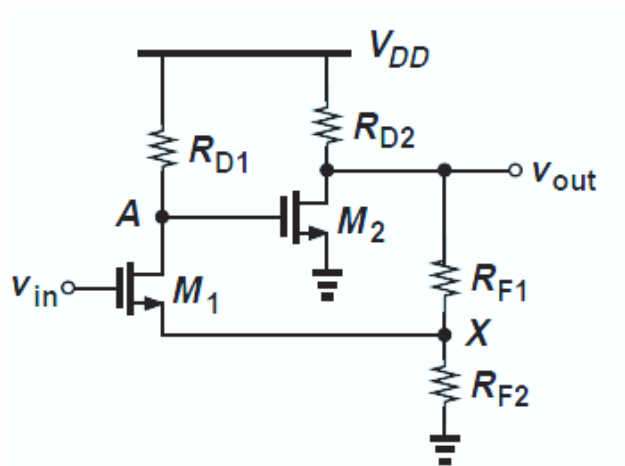


Fig. 4B

- 4C) Explain the working principle of the Colpits Oscillator with a neat circuit diagram. Also, obtain an expression for frequency of oscillation and find the frequency of oscillation for $C_1 = C_2 = 10\mu\text{F}$ and $L = 1\text{mH}$. (3)
- 5A) Determine the closed-loop gain, closed-loop input impedance, and closed-loop output impedances for the circuit shown in Fig. 5A. Consider I/O effects (Loading Effect). Assume $\lambda=0$ for both M_1 and M_2 . Assuming R_M is not small, and the laser diode exhibits an impedance of R_L .

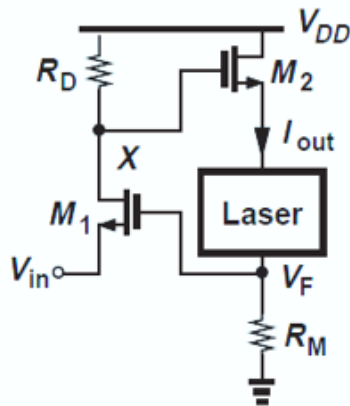


Fig. 5A

- 5B) Determine the expression for % efficiency in Class - A Push - Pull power amplifier and also find its maximum efficiency. (3)
- 5C) Explain the working principle of ring oscillator with a neat circuit diagram. Obtain the expressions for minimum gain and frequency of oscillation. Also, find the frequency of oscillation for drain resistance and capacitance of $20\text{K}\Omega$ and $10\mu\text{F}$. (3)

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